

# COOLCORE WORKWEAR

## SMART WEAR WITH CENTRALIZED COOLING

Adaptive Water-Cooling Smart Workwear is an intelligent protective garment designed for high-temperature industrial environments. Integrating micro water-channel cooling, biosensing modules, and a mobile interaction system, it enables real-time thermal regulation and health monitoring, transforming traditional workwear into a responsive safety solution for heat-stress prevention.

# INSPIRATION








Every summer when I go out, I feel like I'm in a huge steamer, and soon I sweat confusely...

Jiangsu Province is located in the mid-latitude zone of the east coast of the Asian continent, which belongs to the East Asian monsoon climate zone, and is in the climate transition zone between subtropical and warm temperate zones. In summer, the average climate in some areas can reach more than 38 degrees, and the body feels more than 40 degrees.



- How did they endure such heat?

# BACKGROUND

**1600°C** **70°C** **100°C** **50°C**





Ironworkers are skilled laborers who work in iron factories. Their daily routine involves operating heavy machinery and equipment to produce iron products. They handle raw materials, monitor production processes, and ensure safety standards are met. Physical strength and precision are crucial in their job. Despite the challenges, they contribute to building the foundationof modern infrastructure.

Before pouring the steel ladle filled with molten steel, workers need to measure the temperature of the molten steel (approximately 1600°C) in an ambient temperature of about 70°C. Workers in the tundish area are responsible for controlling the flow rate of the tundish and adding protective slag to the mold, which exposes them to prolonged baking at around 100°C. In the billet delivery span, the working environment temperature is approximately 50°C.

# BRAIN STORM









**PORTABLE**

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# QUESTIONNAIRE

### FEELINGS ABOUT SUMMER IN THE LIVING AREA

21.82% 30.3% 33.94% 13.94%

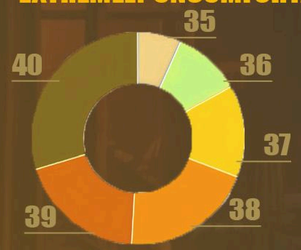
- VERY COOL
- VERY SUNNY AND MUGGY
- A BIT HOT BUT BEARABLE
- HOT AND DRY

### THE MOST COMFORTABLE BODY TEMPERATURE

80.61% 19.39%

- 22-25
- 26-30

### HOW MANY DEGREES WILL MAKE YOU FEEL EXTREMELY UNCOMFORTABLE?




### WHAT FEATURES DO YOU THINK ARE NEEDED?

PERSISTENT COOLING
AIR CIRCULATION
LOW ENERGY CONSUMPTION
HEAT RESISTANCE
LIGHTWEIGHT AND EASY TO PUT ON AND TAKE OFF.

# INTERVIEW




**XIAOWEN**
25, Male

### GRADUATE STUDENT IN CS

It is essential to create a cooling wearable work uniform for ironworks plant workers. We should pay more attention to those working in special environments and utilize new technologies to provide them with better support in their jobs.

### DIGITAL MEDIA STUDENT

Wow! Are we really going to make a cooling garment? It feels absolutely necessary. I think the cooling effect and air circulation are quite important, and it also needs to be heat-resistant in certain areas.



**XIAOCONG**
23, Female

# PERSONA



**CHACEE** 20 years old
Market student
have a study on clothes

I believe that clothing today should not only be created with fashion in mind, but also pay more attention to groups that may be overlooked yet are in urgent need of functional clothing.

### COMCERNS

- The temperature in the ironworks is extremely high
- The workwear is heavy, stuffy, and hot

### EXCEPTIONS

- With effective cooling,
- Air circulation,
- The integration of new technologies.

# PRODUCT ANALYSIS



## NECKBAND FAN

**PROS**
Portable
Easy to Use
Safe

**CONS**
Short battery life
average cooling performance



## ICE COOLING VEST

**PROS**
Highly efficient cooling with wide coverage

**CONS**
Short heat retention time



## AIR-COOLED JACKET

**PROS**
Air Circulation
Uniform Cooling

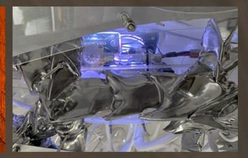
**CONS**
Suboptimal cooling performance

### SUMMARY

It is essential to create a cooling wearable work uniform for ironworks plant workers. We should pay more attention to those work-

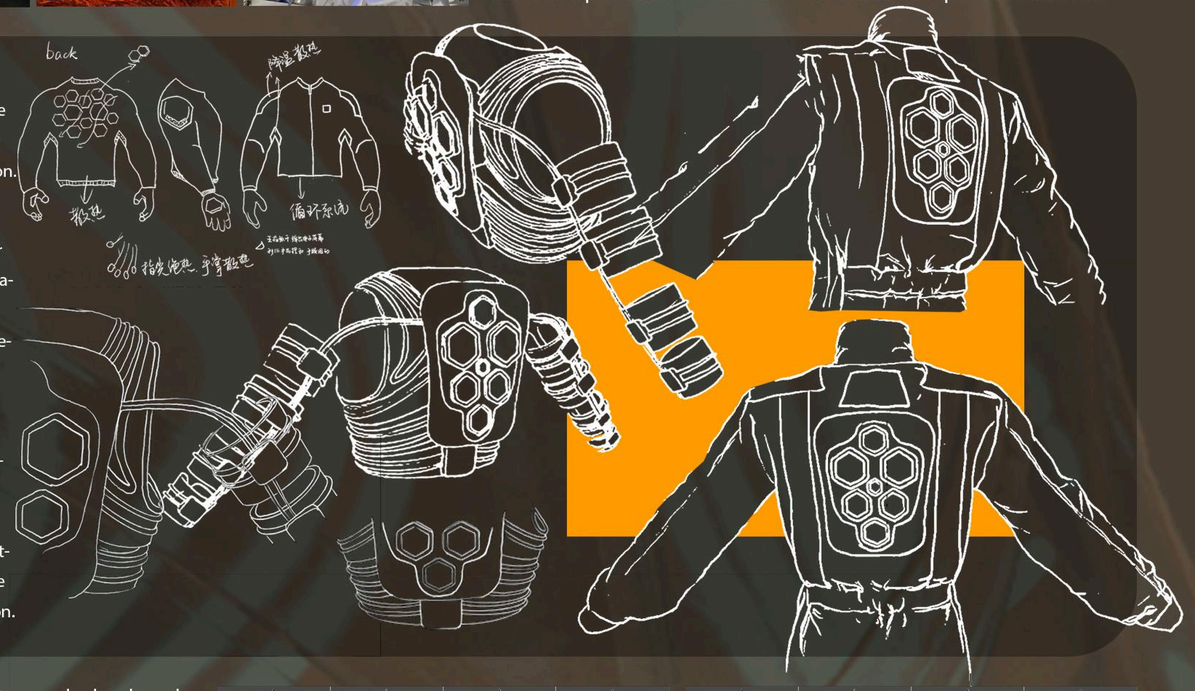
# SKETCH



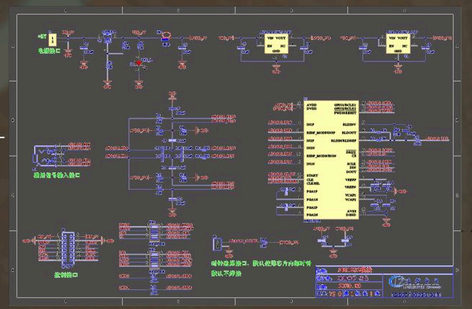
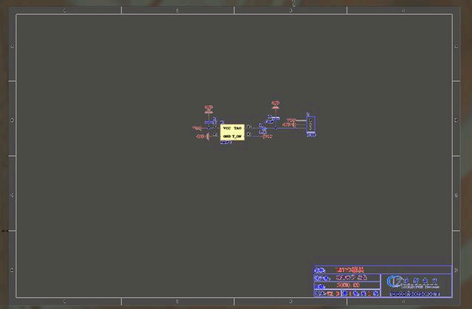


Inspired by the extreme heat of steel mills—scorching metal, rising heat waves, and sweat-soaked workwear. The color palette transitions from rust-red to cooling blue, blending carbon fiber textures with modular cooling structures, exploring how functional wear can become a poetic armor for laborers. reduce occupational hazards.

- Military-grade elastic composite materials ensure flexible protection.
- It enables 8-hour continuous operation, preventing heat stress and reducing occupational hazards.
- The back-mounted hub controls the arm cooling tubes with adjustable temperature and flow direction.

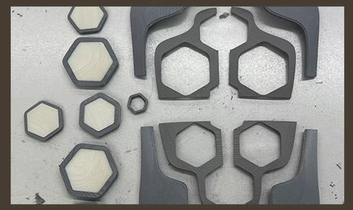
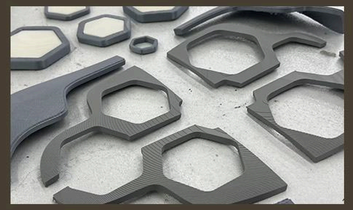
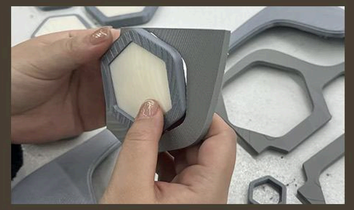


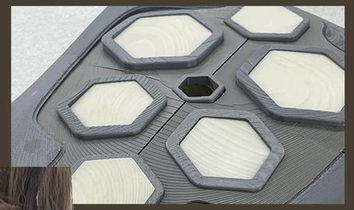
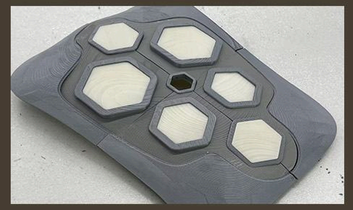

Our device integrates newly developed sensors:
The ADS1292 is a precision analog front-end that amplifies and digitizes bioelectrical signals (e.g., ECG) for health monitoring.
The LMT70 is a high-accuracy analog temperature sensor utilizing semiconductor voltage-temperature characteristics for real-time wearable tracking.

# PROTOTYPE





# FINAL DESIGN





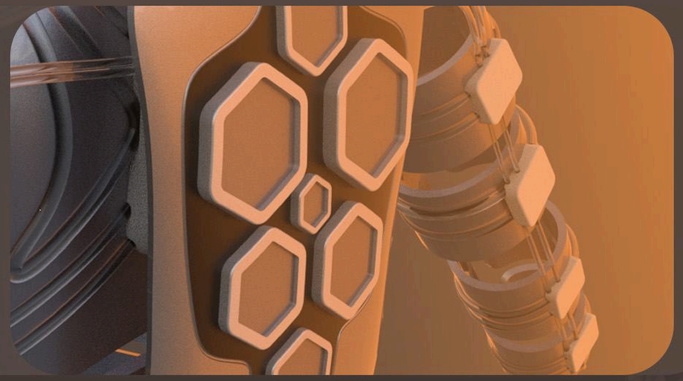


## "Smart Liquid Cooling Workwear"

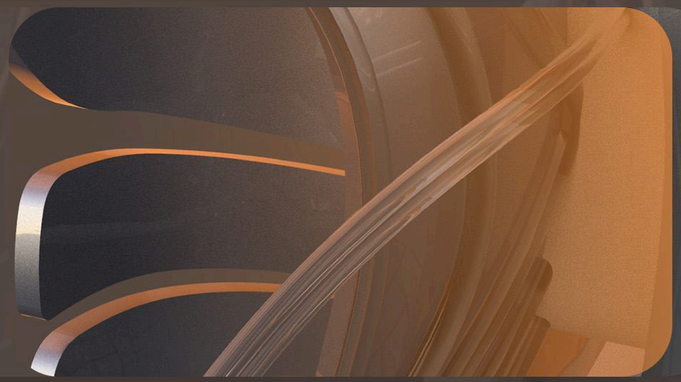
Designed for high-temperature ironmaking environments, it utilizes a flexible micro-tube liquid cooling system and adaptive temperature control algorithms to achieve dynamic body surface cooling through biomimetic microchannels embedded in the garment lining.

Military-grade elastic composite materials ensure flexible protection, with 8-hour continuous operation to effectively prevent heat stress injuries and reduce occupational hazards. The arm cooling tubes are regulated by a back-mounted control hub, allowing adjustable water temperature and flow direction.

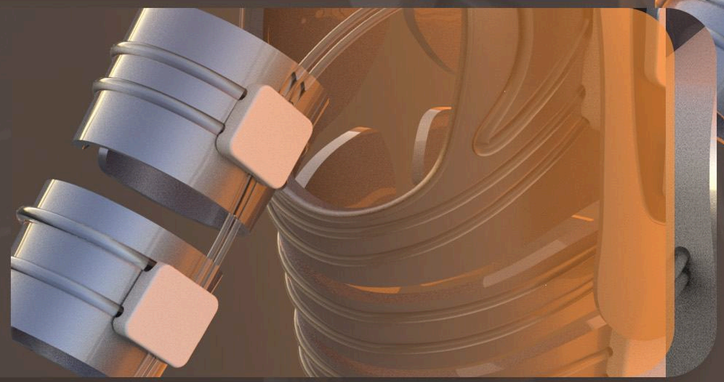
The back-mounted control hub consists of six hexagonal modules, regulating water flow rate, direction, temperature, air content, and more. Under its command, the branch controllers on the arms receive signals to precisely optimize cooling performance across every cooling tube



Flexible liquid-cooling tubes are strategically woven throughout the garment's inner layer, following the body's natural perspiration patterns for optimal cooling. The heat-resistant tubing resists deformation, enhancing durability while maintaining a comfortable working environment.



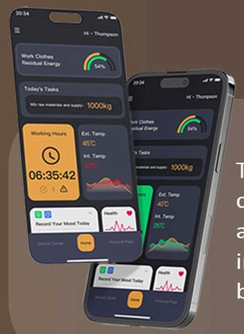
Upon receiving commands from the central control hub, the arm-mounted branch controllers precisely regulate four intersecting cooling tubes - adjusting flow rate, direction and temperature in real-time. This ensures synchronized, high-performance cooling across the entire liquid-cooling network.





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## UI DESIGN



The main page displays your continuous working hours, and if it turns orange, that indicates it's time to take a break.



The app features pages for health monitoring, personal profiles, device detection, and work requests, meeting all on-site worker needs with real-time health and equipment monitoring.

## SERVICE BLUEPRINT

| TIME | Awareness | Acquisition | Utilization | Maintenance | Replacement |
|---|---|---|---|---|---|
| CUSTOMER ACTION | Participated in corporate /learn from colleagues<br>Saw corporate promotional materials | Submit request via internal system/app<br>Register & collect at distribution point | power<br>parts<br>temp<br>hours | Detect anomaly Report anomaly<br>Routine cleaning & maintenance | Submit request<br>Return/Retrieve |
| TOUCHPOINT | Trainer intro /Peer sharing<br>Poster, internal notice, product page | Internal system/app Supply station | APP | | |
| EMPLOYEE ACTION | Clear product presentation | Verify → Confirm → Distribute → Instruct | Admin patrol App backend monitoring | | MAINT: Check/Repair CLEAN: Periodic wash |
| TECHNOLOGY | Training System \| Promotional Design | Internal System \| Identity Recognition Tech | Sensors × App × Wireless Comms | Fault Diagnosis \| Remote O&M | Asset recovery Eco-label certification |
| BACKSTAGE EMPLOYEE | Promotional Material Production \| System & Tech Support \| Supply Chain & Recycling | | | | |